



Project Close-out Report

(Dead-man-switch) Permission based Wallet, shipped as Epora

Name of project	(Dead-man-switch) Permission based Wallet, shipped as Epora
Project URL	projectcatalyst.io , Fund 11, Cardano Use Cases: Concept
Project number	1100002
Project manager	Sandro Schaier
Date project started	11 March 2024
Date project completed	[To fill: date of the final milestone submission]
Plan	7 months, with deliveries from April to October 2024
Budget	100,000 ADA in five milestones; 59,000 ADA distributed by 25 September 2026
Close-out video	[To fill: YouTube or Vimeo link]

1 Challenge KPIs and how the project addressed them

The challenge brief states aims, not numeric KPIs, so each row quotes one aim and the project's answer.

Aim in the challenge brief	How the project addressed it
"deliver proof of concept, design research and basic prototyping"	The whitepaper v1.0 gives the design, a formal model, and a security analysis. The Aiken validators and the reference app run on Preprod at epora.io . Mainnet beta: [To fill: mainnet URL] .
"generate and test new ideas that bring utility to Cardano"	The Preprod phase produced 49 tracked issues, and 2 of them came from outside testers. Each issue has one category, and 46 are fixed.
"produce specifications and a pathway that requires more significant technical development"	The whitepaper and the OpenAPI 3.1 document specify the system. Section 5 gives the pathway: a smart-wallet CIP, an external audit, and wallet integrations.
"for the benefit of the Cardano ecosystem"	The project's own code and documents are public under the MIT license. Third-party parts keep their own licenses.

2 Project KPIs and how the project addressed them

The proposal measures success through its five milestones. The table summarizes the acceptance criteria from the Statement of Milestones, the result, and the review status in the Milestone Module on 25 September 2026.

Milestone	Acceptance criteria	Result	Status
1. Setup, whitepaper, planning	X and Discord accounts; a whitepaper with abstract, introduction, functionality, permission system, and use cases; a development task list	X account @schaier_io and a Discord server; whitepaper v1.0 with its LaTeX source; a task tracker with one file per milestone	Approved 7 Jul 2026
2. Smart contract development	The whitepaper methods; code documentation for new developers; abuse vectors explained and shown not to work	Aiken validators for every whitepaper transition; a contracts guide; a security evidence map ; 750 of 750 contract tests pass	Approved 6 Aug 2026
3. Development of UI	A video of the whitepaper functions; an API and a specification for external developers; code documentation	A wallet UI video; a public v1 API with an OpenAPI 3.1 document and a developer guide ; the app README	Approved 21 Sep 2026
4. Testnet launch	A usable prototype; aggregated feedback; improvements and bug fixes from the feedback	The prototype on Preprod at epora.io ; a feedback page with 49 categorized issues; 46 fixes with linked pull requests	Submitted, in review

Milestone	Acceptance criteria	Result	Status
5. Finalization and mainnet launch	A mainnet prototype; categorized feedback; communication with potential wallet developers, if they or the community ask for it; this report; a close-out video	Mainnet beta at [To fill: URL] ; the same feedback page; a wallet outreach page ; this report; the video in Section 8	This submission

Still open. WalletConnect signing (pairing works), a screen for governance votes, the retest of [#561](#), and the results of the wallet talks. See the [task tracker](#).

Timeline and scope. The plan had seven months, with the last delivery in October 2024. The Milestone Module approved milestones 1 to 3 between 7 July and 21 September 2026. The project delivered much more than the proposal asked for, and the budget stayed at 100,000 ADA. The larger scope took more time than the plan allowed.

Proposal	Delivered
Permissions after a period of inactivity Two parties agree before a transaction	A proof-of-life dead-man-switch that owners and liveness keepers renew Weighted multi-signature for operator actions, with a workspace where co-signers review and sign
Limits for spending certain assets A family member recovers funds after six inactive months Partially custodial wallets	A per-day allowance for each user, with up to 5 assets in each allowance Weighted-share recovery for several beneficiaries, each with an optional unlock delay, and exact distribution to all of them Third parties can hold a role: co-owner, spender, liveness keeper, or beneficiary. The funds stay in the contract.
<i>Not in the proposal</i>	Up to 15 streaming payments that settle while the owner is offline; reward withdrawal, certificates, and governance votes from the smart wallet; stake-credential pinning and consolidation; a formal model and a security analysis; 750 contract tests, 37 of them property-based

3 Key achievements

- **Categorized feedback.** Reports came from team test sessions, from GitHub issue forms linked in the app footer, and from Discord. All 49 tracked issues have one category: 46 fixed, 1 for the next version, and 2 not a bug.
- **Outside reports led to changes.** A GitHub report from Isaac (bytegen-dev), made with the Vespr wallet, led to a consolidation change ([#562](#)). A Discord request from Bee (opacular) led to a co-signer invite link and a signed-in status for each co-signer ([#568](#) to [#571](#)).
- **Wallet developer outreach.** A public message in the Eternl Discord on 24 September 2026 asked about a native Eternl integration and a CIP for a standard smart wallet. Eternl replied by direct message (*reported*). We sent Begin Wallet a direct message and contacted other wallet developers (*reported*).
- **Mainnet preparation.** Network configuration, beta consent, legal pages, and a network switch merged on 25 September 2026 ([#589](#), [#590](#), [#594](#)).

4 Key learnings



One action in Epora: two contracts read the same redeemer, and a normal wallet signs and pays.

- **Two contracts must agree on one action.** Both contracts read one redeemer, the declared action. The state contract proves that the action equals the true change to the wallet state. The wallet contract limits the payout to that action and has no permission logic, so a reviewer can check it by itself.

- **One configurable contract replaces many bespoke ones.** Six features give 364 combinations of “and” and “or”, and seven give 1,093. One contract for each combination is too much to build, test, and audit. So the whole configuration is one datum on one contract, and a new combination is data, not code.
- **An open deposit address has edge cases.** Anyone can pay into the wallet with no datum, as into a normal address. But a sender can attach a datum hash, or pick a foreign stake credential that then collects the staking rewards of those funds. The contracts now pin the stake credential and reject datum hashes on outputs that stay in the wallet. The app offers to consolidate stray funds. An earlier whitepaper version missed the datum-hash case, and the whitepaper now records the correction.
- **Ledger limits set the size of the product.** Every check must fit the size limit and the execution budget of one transaction. So the contracts cap the lists: 15 user and beneficiary records (at most 10 users), 15 streaming payments, and 5 assets in one allowance. The beneficiary checks compare every pair, so their cost grows with the square of the list. A reference script keeps spends small, and the app offers a preparation step when a recovery does not fit.
- **A smart wallet still needs a normal wallet.** Today each action uses two wallets. A normal CIP-30 wallet signs, pays the fee, provides the collateral, and gets the change. CIP-30 does not cover script wallets, so the app must show the funds in the smart wallet and preview each transaction. Native wallet support would remove the second wallet. This is the aim of the CIP in Section 5.

5 Next steps for the product

- **Mainnet beta.** Run the beta with small funds, and close the open items from Section 2. On Mainnet, a notice says that the code has no external audit, and users cannot close it.
- **A smart-wallet CIP.** EVM chains have smart-account standards, for example ERC-4337 and ERC-7579. We plan to draft a similar CIP for Cardano. It would define how a wallet finds a smart wallet, shows its funds and rules, and builds its actions. It can build on CIP-141 (Plutus wallets) and CIP-106 (multisig wallets), which both have the status Proposed. After an external audit, the Epora contracts could be the reference implementation.
- **Wallet integrations.** Follow up on the Eternl reply, the Begin Wallet contact, and other wallet teams.
- **Funding and commercial use.** **[To fill: further funding plans and commercial use, if any]**

6 Final thoughts

The proposal addressed the risk of losing all funds when a key or a backup is lost. The delivered contracts, app, and API give Cardano users recovery and shared permissions without a custodian of funds. We want this work to become the base of a smart-wallet standard for Cardano. Feedback stays open on GitHub and Discord.

7 Links to other project sources

Documents	Repository ; whitepaper ; security evidence map ; API guide ; live API reference
Progress	Task tracker ; categorized feedback ; wallet outreach
Prototypes	Preprod at epora.io ; mainnet at [To fill: mainnet URL]
Catalyst and community	Milestone Module ; Discord discord.gg/2uh4BynQBW ; X @schaier_io

8 Close-out video

[To fill: public YouTube or Vimeo link, 2 to 5 minutes, 720p or 1080p]

Sources. Catalyst project page and Milestone Module (read on 25 September 2026); linked repository files; aiken check, Aiken v1.1.23 (750 passed, 0 failed, same day). Items marked *(reported)* have no public record.